

Convolution applications:

- Computer vision
- AI for graphics

□ A BI system analyzes a small greyscale icon patch with a local contrast filter.

a) Given

$$X = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 2 & 1 & 4 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

compute the top-left output entry $Y_{1,1}$.

⇒ To get $Y_{1,1}$ in a **valid** convolution, we take the top-left 2×2 submatrix and apply filter K .

$$X_{1:2,1:2} = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \quad K = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$Y_{1,1} = 1 \cdot 1 + 2 \cdot (-1) + 0 \cdot (-1) + 3 \cdot 1$$

$$= 1 - 2 + 0 + 3$$

$$\boxed{Y_{1,1} = 2} \quad \checkmark$$

b) Compute the full valid convolution output matrix Y

$$\Rightarrow Y_{1,1} = 2$$

$$\Rightarrow Y_{1,2} = \begin{bmatrix} 2 & 0 \\ 3 & 1 \end{bmatrix} \odot \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$= 2 \cdot 1 + 0 \cdot (-1) + 3 \cdot (-1) + 1 \cdot 1$$

$$\boxed{Y_{1,2} = 0} \quad \checkmark$$

$$\Rightarrow Y_{2,1} = \begin{vmatrix} 0 & 3 \\ 2 & 1 \end{vmatrix} \odot \begin{vmatrix} 1 & -1 \\ -1 & 1 \end{vmatrix} \\ = 0 \cdot (1) + 3 \cdot (-1) + 2 \cdot (-1) + 1 \cdot 1 \quad \boxed{= -4} \quad \checkmark$$

$$\Rightarrow Y_{2,2} = \begin{vmatrix} 3 & 1 \\ 1 & 4 \end{vmatrix} \odot \begin{vmatrix} 1 & -1 \\ -1 & 1 \end{vmatrix} \\ = 3 \cdot 1 + 1 \cdot (-1) + 1 \cdot (-1) + 4 \cdot 1 \quad \boxed{= 5} \quad \checkmark$$

$$Y = \begin{vmatrix} Y_{1,1} & Y_{1,2} \\ Y_{2,1} & Y_{2,2} \end{vmatrix} = \begin{vmatrix} 2 & 0 \\ -4 & 5 \end{vmatrix} \quad \checkmark$$

c) Determine the position of the largest response and explain what this says about the local pattern in the input.

$\Rightarrow$ The position of the largest response is at $\boxed{Y_{2,2} = 5}$

$\Rightarrow$ Since we have a diagonal contrast filter, it gives a high positive value when:

- The values on the main diagonal are high (3 and 4).
- The values on the anti-diagonal are low.

$$Y_{2,2} \Rightarrow \begin{vmatrix} 3 & 1 \\ 1 & 4 \end{vmatrix} \quad)$$

2 A dashboard image is processed with a vertical edge detector.

a) Given

$$X = \begin{vmatrix} 1 & 1 & 1 & 4 \\ 1 & 1 & 1 & 4 \\ 1 & 1 & 1 & 4 \\ 1 & 1 & 1 & 4 \end{vmatrix}, \quad K = \begin{vmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{vmatrix},$$

compute $Y_{1,1}$, $Y_{1,2}$,

$\Rightarrow Y_{1,1}$ corresponds to the top-left submatrix

$$X_{1:3, 1:3} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}$$

$$\Rightarrow Y_{1,1} = 1 \cdot 1 + 1 \cdot 0 + 1 \cdot (-1) = 0 \quad \leftarrow \text{every row}$$

$$\boxed{Y_{1,1} = 0} \quad \checkmark$$

$\Rightarrow$ We then slide the kernel over one column to the right to get $Y_{1,2}$.

$$\Rightarrow X_{1:3, 2:4} = \begin{bmatrix} 1 & 1 & 4 \\ 1 & 1 & 4 \\ 1 & 1 & 4 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}$$

$$\Rightarrow \text{Each row: } 1 \cdot 1 + 1 \cdot 0 + 4 \cdot (-1) = -3$$

$$\boxed{Y_{1,2} = -3 - 3 - 3 = -9} \quad \checkmark$$

b) Compute the full valid feature map Y .

$\Rightarrow$ Since we are sliding a 3×3 kernel over a 4×4 input matrix, the output dimensions for the valid convolution feature map are:

$$\begin{aligned} \text{output.d} &= (d_x - d_k + 1) \\ &= (4 - 3 + 1) = 2 \quad \checkmark \end{aligned}$$

$\Rightarrow Y$ is a 2×2 matrix.

$$\Rightarrow Y_{1,1} = 0 \quad \checkmark$$

$$\Rightarrow Y_{1,2} = -9 \quad \checkmark$$

$$\Rightarrow Y_{2,1} \Rightarrow \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \quad \Rightarrow \text{This is identical to the calc. for } Y_{1,1}$$

$$\boxed{Y_{2,1} = 0} \quad \checkmark$$

$$\Rightarrow Y_{2,2} \Rightarrow \begin{vmatrix} 1 & 1 & 4 \\ 1 & 1 & 4 \\ 1 & 1 & 4 \end{vmatrix} \Rightarrow Y_{3,2} = -9 \quad \text{Identical to calc. for } Y_{1,2}!$$

$$Y = \begin{vmatrix} 0 & -9 \\ 0 & -9 \end{vmatrix} \quad \checkmark$$

c) Explain what the sign of the entries means and state where the vertical edge is located.

$\Rightarrow$ The sign indicates the direction of intensity change (gradient) relative to the filter's location.

$\Rightarrow$ In the filter K , the right column has negative weights (-1 entries) and the left column has positive weights (+1).

$\Rightarrow$ A negative result means that the pixels on the right side of the image are much brighter (more intense) than those on the left.

$\Rightarrow$ Location of vertical edge: the edge is located in the transition from column 3 to column 4.

Reading $\Rightarrow$ Google: YOLO OpenCV

5] A greyscale patch is processed with a Laplacian filter using same padding.

a) Given

$$X = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad K = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix},$$

write the zero-padded matrix $\tilde{X}$

$\Rightarrow$ To achieve same-padding for the 3×3 input X , with a 3×3 filter, we use a border of 1 pixel of zeros on all sides

$$\tilde{X} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

b) Compute $Y_{1,1}$, $Y_{2,2}$, $Y_{3,3}$

$$\Rightarrow Y_{1,1} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix} \odot \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix} = \begin{aligned} & 0 \cdot 0 + 0 \cdot (-1) + 0 \cdot 0 \\ & + 0 \cdot (-1) + 0 \cdot 4 + 1 \cdot (-1) \\ & + 0 \cdot 0 + 1 \cdot (-1) + 2 \cdot (0) \end{aligned}$$

$$Y_{1,1} = -1 - 1 + 0 = -2 \quad \checkmark$$

$$\Rightarrow Y_{2,2} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 0 \end{bmatrix} \odot \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix} = \begin{aligned} & 0 \cdot 0 + 1 \cdot (-1) + 0 \cdot 0 \\ & + 1 \cdot (-1) + 2 \cdot 4 + 1 \cdot (-1) \\ & + 0 \cdot 0 + 1 \cdot (-1) + 0 \cdot 0 \end{aligned}$$

$$Y_{2,2} = -1 + 6 - 1 = 4 \quad \checkmark$$

$$\Rightarrow Y_{3,3} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \odot \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix} = \begin{aligned} & 2 \cdot 0 + 1 \cdot (-1) + 0 \cdot 0 \\ & + 1 \cdot (-1) + 0 \cdot 4 + 0 \cdot (-1) \\ & + 0 \cdot 0 + 0 \cdot (-1) + 0 \cdot 0 \end{aligned}$$

$$Y_{3,3} = 0 - 1 - 1 + 0 = -2 \quad \checkmark$$

c) Compute the full same-padded output Y , and identify the position with the strongest positive response.

⇒ We already have $Y_{1,1}$, $Y_{2,2}$, $Y_{3,3}$

⇒ We will use the property of the Laplacian filter that is the sum-of-neighbors comparison

$$K = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{bmatrix} \Rightarrow \text{The center (4) is the multiplier for the "target" pixel.}$$

⇒ The neighbors of the "target" pixel are subtracted (-1).

$$\Rightarrow Y_{1,2}: \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 2 & 1 \end{bmatrix}, \text{ Target: } 1, \text{ neighbors: } (0, 0, 0, 2)$$

$$Y_{1,2} = 1 \cdot 4 - (0 + 0 + 0 + 2) = 2 \quad \checkmark$$

$$\Rightarrow Y_{1,3}: \begin{bmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 2 & 1 & 0 \end{bmatrix}, \text{ Target: } 0, \text{ neighbors: } (0, 1, 0, 1)$$

$$Y_{1,3} = 0 \cdot 4 - (0 + 1 + 0 + 1) = -2$$

$$\Rightarrow Y_{2,1}: \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}, \text{ Target: } 1, \text{ neighbors: } (0, 0, 2, 0)$$

$$Y_{2,1} = 1 \cdot 4 - (0 + 0 + 2 + 0) = 2$$

$$\Rightarrow Y_{2,3}: \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \text{ Target: } 1, \text{ neighbors: } (0, 2, 0, 0)$$

$$Y_{2,3} = 1 \cdot 4 - (0 + 2 + 0 + 0) = 2$$

$$\Rightarrow Y_{3,1}: \begin{bmatrix} 0 & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \text{ Target: } 0, \text{ neighbors: } (1, 0, 1, 0)$$

$$Y_{3,1} = 0 \cdot 4 - (1 + 0 + 1 + 0) = -2$$

$$\Rightarrow Y_{3,2} : \begin{vmatrix} 1 & 2 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{vmatrix}, \text{ Target: } 1, \text{ neighbors: } (2, 0, 0, 0)$$

$$Y_{3,2} = 1 \cdot 4 - (2 + 0 + 0 + 0) = 2$$

$$Y = \begin{vmatrix} -2 & 2 & -2 \\ 2 & 4 & 2 \\ -2 & 2 & -2 \end{vmatrix} \quad \checkmark$$

$\Rightarrow$ The strongest positive response is 4, at position $Y_{2,2}$!

$\Rightarrow$ The high positive value indicates a spike or bright spot.

$\Rightarrow$ The positive neighbors (2s) are ridges around the spot

$\Rightarrow$ The negative entries at the corners (-2s) represent steep drop-off in intensity.

10 An RGB patch is processed with one $2 \times 2 \times 3$ filter.

a) The RGB channels are

$$R = \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} \quad G = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} \quad B = \begin{vmatrix} 2 & 0 \\ 1 & 1 \end{vmatrix}$$

the filter channels:

$$K_R = \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} \quad K_G = \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} \quad K_B = \begin{vmatrix} -1 & 0 \\ 0 & -1 \end{vmatrix}$$

Compute the contribution of the R-channel to the output scalar.

$\Rightarrow$ Since the filter is 2×2 and the input is 2×2 , we only have one filter position. $\Rightarrow$ No sliding.

$$R \Rightarrow \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} \odot \begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 \cdot 1 + 2 \cdot 0 + 0 \cdot 0 + 1 \cdot 1 \\ = 1 + 0 + 0 + 1$$

$$\begin{array}{c} \text{Contr-}R \\ \hline = 2 \end{array} \quad \checkmark$$

b) Compute the G and B channel contributions.

$$G \Rightarrow \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = 0 \cdot 0 + 1 \cdot 1 + 1 \cdot 1 + 0 \cdot 0$$

Contr_G

$$= 2$$

$$B \Rightarrow \begin{vmatrix} 2 & 0 \\ 1 & 1 \end{vmatrix} \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = 2 \cdot (-1) + 0 \cdot 0 + 1 \cdot 0 + 1 \cdot (-1)$$

Contr_B

$$= -3$$

c) Compute the final output scalar and interpret whether the patch is more red-green dominant or more blue dominant with respect to this filter.

$\Rightarrow$ To get the final output scalar, we sum the contributions of all 3 channels:

$$Y = \text{Contr}_R + \text{Contr}_G + \text{Contr}_B$$

$$Y = 2 + 2 - 3 = 1$$

$\Rightarrow$ The patch is clearly red-green dominant, because the contributions from R and G are positive, while that of B is negative.

11) A BI team segments four customers with $K=2$.

a) Given the points

$A(1,1)$, $B(1,2)$, $C(5,4)$, $D(6,5)$,

and initial centroids

$$m_1^{(0)} = (0,0), \quad m_2^{(0)} = (6,4),$$

compute the distances from each point to both centroids, and assign each point to a cluster.

- Rules:
- Use squared Euclidean distances
 - Break ties for smaller cluster idx
 - Update centroids at least once

Point	$d^2(x, m_1)$	$d^2(x, m_2)$	Winner
$A(1,1)$	$(1-0)^2 + (1-0)^2 = 2$	$(1-6)^2 + (1-4)^2 = 34$	C_1
$B(1,2)$	$(1-0)^2 + (2-0)^2 = 5$	$(1-6)^2 + (2-4)^2 = 29$	C_1
$C(5,4)$	$(5-0)^2 + (4-0)^2 = 41$	$(5-6)^2 + (4-4)^2 = 1$	C_2
$D(6,5)$	$(6-0)^2 + (5-0)^2 = 61$	$(6-6)^2 + (5-4)^2 = 1$	C_2

Assignments $\Rightarrow$

$C_1: \{A, B\}$
 $C_2: \{C, D\}$

b) Compute the updated centroids

$$m_1^{(1)}, \quad m_2^{(1)}.$$

$\Rightarrow$ To get the new centroids, we take the arithmetic mean of all points in a cluster:

$$\frac{1}{|C_k|} \sum_{x \in C_k} x$$

$$C_1 \Rightarrow A(1,1), B(1,2) \Rightarrow x = \frac{1+1}{2} = 1 \checkmark$$

$$y = \frac{1+2}{2} = 1.5 \checkmark$$

$$\Rightarrow m_1^{(1)} = (1, 1.5)$$

$$C_2 \Rightarrow C(5,4), D(6,5) \Rightarrow x = \frac{5+6}{2} = 5.5$$

$$y = \frac{4+5}{2} = 4.5$$

$$\Rightarrow m_2^{(1)} = (5.5, 4.5)$$



c) Perform one more assignment step with the updated centroids and decide whether the algorithm has converged.

Point	$d^2(x, m_1)$	$d^2(x, m_2)$	Winner
A(1,1)	$(1-1)^2 + (1-1.5)^2 = 0.25$	$(1-5.5)^2 + (1-4.5)^2 = 32.5$	C_1
B(1,2)	$(1-1)^2 + (2-1.5)^2 = 0.25$	$(1-5.5)^2 + (2-4.5)^2 = 26.5$	C_1
C(5,4)	$(5-1)^2 + (4-1.5)^2 = 22.25$	$(5-5.5)^2 + (4-4.5)^2 = 0.5$	C_2
D(6,5)	$(6-1)^2 + (5-1.5)^2 = 37.25$	$(6-5.5)^2 + (5-4.5)^2 = 0.5$	C_2

Assignments $\Rightarrow$
 $C_1: \{A, B\}$
 $C_2: \{C, D\}$

$\Rightarrow$ Convergence!